Consequence-Aware Failure Diagnostics for Learned Particle-Reconstruction Representations

Collaboration concept note

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Abstract

A monitor on a deployed reconstruction model must answer two questions that average accuracy does not: which declared mechanism caused an alarm, and what scientific consequence followed. We propose a replay-based diagnosis of frozen representations under controlled interventions, organised as three tiers with one direction of travel — mechanism, realism, real data. A controlled waveform simulator in which both the assumed covariance $\hat{\Sigma}$ entering the training objective and the realized covariance Σ are known supplies the mechanism tier: there we test the prediction that an unweighted representation displacement is the wrong-metric statistic — the consequence-relevant quantity is displacement in the Σ^{-1}-whitened metric pulled back through the output Jacobian — using perturbation families that dissociate alarm from consequence by construction. A paired neutrino-telescope production (one seeded injection replayed through two detector geometries, with content-matched acquisition/-physics cells and undeclared abstention families) supplies the realism tier, testing the mechanism tier's failure predictions out of sample on a frozen public model. The public TIDMAD benchmark supplies the real-data check: one compact transformer trained under two covariance assumptions on identical data. Layerwise evidence is compared against input tests, MMD, classifier two-sample tests, embedding distance, uncertainty, and output-only monitoring at a matched false-alert budget, with abstention on families the study never declared.

1 Gap and contribution

Existing shift monitors can detect changes in inputs, outputs, or embeddings, but a scientific reconstruction pipeline requires two additional distinctions: whether the change reflects corrupted acquisition or valid but under-supported physics, and whether the detected change materially degrades a physics result. We test whether intermediate representations improve this attribution beyond generic monitoring at the same false-alert budget, while allowing abstention on ambiguous or unseen failures. The attribution claim is conditional on the declared simulator and forward model: unknown physics and model misspecification are not identifiable from a response profile alone and are routed to abstention.

No public dataset can decide these questions. None carries a known assumed-versus-realized covariance, and none is event-paired across the factors a causal claim must vary: the released NuBench sets are independent per-geometry productions, and TIDMAD's own authors disclaim the scientific meaning of its denoising score. The study therefore produces its own evidence in two designed datasets — a controlled-covariance waveform set (**ORACLE-Cov**) and a paired multi-geometry point-cloud set (**ORACLE-Paired**), the latter a citable artifact in its own right — and uses TIDMAD as the real-data check. The tiers are ordered: predictions fixed on the mechanism tier are pre-registered before the single confirmatory run on the realism tier.

In waveform reconstruction, structured acquisition noise induces a likelihood geometry through the inverse covariance: the correct residual weighting is Σ^{-1}, and a model evaluated under the wrong weighting can prefer reconstructions that are worse in the metric the instrument implies. Architecture and training coverage further determine which physically relevant distinctions survive into the representation [17]. This study asks whether violations of these assumptions can be diagnosed after training, and whether a detected violation is scientifically consequential. The question has become concrete: cross-modality frozen backbones now exist — Panda V2 is pre-trained with one recipe on LArTPC, collider-TPC and water-Cherenkov data and adapted with 10^3 labels per detector [18] — and their authors list transfer across detectors and from simulation to real data as untested. Those are exactly the acquisition-shift and support-shift questions posed here, on a frozen public model. The artifact is a reproducible experiment suite—configurations, adapters, hooks, replay scripts, and reports—not a maintained software product.

Position relative to learned-geometry reconstruction. The subjects of this study are models that *learn* the geometry of the measurement: distance-weighted graph networks that place every sensor in a learned coordinate space [19], potential-based objectives that attach to each vertex a learned weight for whether it can represent its object [20], and cross-detector pre-training recipes [18]. None of these carries a statement of what its representation can resolve once the acquisition or the physics moves; recent benchmarks of physics foundation models report regime-dependent scores across distribution shifts without a mechanism [21]. The learned resolvability weight of object condensation is the closest relative of the Fisher-rank diagnostic of Section 3, which computes the same quantity for a *frozen* model from its Jacobian and the assumed covariance, with no training signal. Nothing here proposes a new reconstruction architecture; the contribution is the contract, the mechanism, and the designed data on which the contract can be falsified.

Identifiability limit. Diagonalizing a covariance yields decorrelated statistical modes, not labeled physical sources: $\Sigma = \Sigma_1 + \Sigma_2$ admits infinitely many positive-semidefinite splits, and low-rank factors are rotation-degenerate. All attribution here is over *declared intervention families* under a stated simulator. Observing a response profile does not identify the cause of an unknown real shift, and we make no such claim.

2 Failure contracts

Three declared labels, frozen before evaluation.

N—acquisition-contract violation. For this study the closed family is power-spectral-density (PSD) or covariance change, jitter, drift, spectral lines, glitches, tails, hit thinning, and channel or module loss. The measurement violates a declared acquisition condition; no broader N claim is made.

S—training-support shift. Window or event composition moves toward physical strata that are sparse under the clean training distribution. The events themselves are clean and physically valid; only the support has moved.

E—evaluation-contract fault. Wrong units, coordinate ordering, output semantics, PSD convention, or metric implementation. This is a pipeline bug, not a distribution shift.

N versus S is the scientific core. E is decided by deterministic schema and range invariants, so it is reported as a gate result, not as a class in a classification score. The informative and difficult separation is N from S. N and S may both move the observed input distribution, so generic input-shift detection alone cannot distinguish them. Evidence for N comes from acquisition-quality indicators, paired corruption interventions, or known violations of the measurement model. S is defined by physically valid, uncorrupted events that occupy low-density regions of the clean training support. The scientific problem is therefore not whether the input changed, but whether the change reflects corrupted measurement or valid but under-supported physics. Section 3 states why the two leave different geometric signatures in a frozen representation, which is what makes N versus S a prediction rather than a label. That distinction is what matters operationally: discarding S rejects physically valid events, whereas rejecting N removes measurements that violate the declared acquisition contract.

3 Mechanism: what a frozen representation can resolve

Write the declared forward model as $x = g(z) + n$, $n \sim \mathcal{N}(0, \Sigma)$, with $z \in \mathbb{R}^m$ the physical latent variables (energy, direction, vertex, multiplicity, …) and $J_g(z) = \partial g/\partial z$. Three first-order statements, each decided on the controlled tier where g, Σ and z are all known, turn the N and S labels into predictions.

Consequence is distance in the pullback metric. For fixed Σ the Fisher information of z is exactly $I(z) = J_g(z)^\top \Sigma^{-1} J_g(z)$: the Gaussian likelihood pulls the Σ^{-1} inner product back onto latent space, and $\delta z^\top I(z)\, \delta z$ measures a latent perturbation in detector-distinguishability units — how far it moves the observation relative to what the noise can resolve [17]. The same construction through a frozen model's output map h gives $J_h^\top \Sigma^{-1} J_h$ on any intermediate representation. An unweighted Euclidean displacement in representation space is therefore the wrong-metric statistic; the consequence-relevant alarm is the displacement in the Σ^{-1}-whitened metric pulled back through the output Jacobian (the *pullback-whitened* statistic of Tier 1), and its Σ-free surrogate is the projection of the displacement onto the leading right-singular directions of the output Jacobian (the *Jacobian-projected* statistic used on Tier 2). The output-null and output-aligned arms of Section 4 are built from the same Jacobian, which is why their predicted sign is a consequence of this metric and not a free parameter.

Support shift lives in the unexcited tangent complement. Training varies only the latent coordinates $S \subseteq \{1, \ldots, m\}$ that the simulation or calibration procedure excites. In the population limit the reconstruction objective supplies first-order identifying information only along $T_S(z) = \mathrm{span}\{\Sigma^{-1/2}\partial g/\partial z_j : j \in S\}$; the complementary directions are never tested by the loss — exactly in the linear case, and locally for nonlinear g by the same Jacobian argument. Whatever the frozen model does along $T_S^\perp$ was selected by its architecture and initialization, not by the data. A support shift — clean events whose latent coordinates move into strata the training never excited — therefore produces a whitened observation displacement $\Sigma^{-1/2}\big(g(z') - g(z)\big)$ whose leading component lies in $T_S^\perp$, where the model's Jacobian carries no trained sensitivity. Where $\operatorname{rank} I(z) < m$ the latent variables are not even locally identifiable at z, for any decoder capacity; strata in which this rank drops are declared abstention regions, not attribution failures.

Acquisition shift lives in the excited span. An acquisition violation changes the noise, $\Sigma \to \Sigma'$ or $\Sigma + \Sigma_{\mathrm{art}}$, and leaves g unchanged. Its signature is a whitened residual whose per-direction variance departs from unity by the spectrum of $\hat{\Sigma}^{-1}\Sigma'$, spread over the directions the training did excite, because that is where the encoder's assumed $\hat{\Sigma}$ was doing work. This is the mechanistic asymmetry the monitors are designed to read: N moves the residual *within* T_S; S moves the representation *along* $T_S^\perp$. A generic input-shift statistic records only that something moved.

Layerwise monitors. For stage maps $r_{\ell+1} = \phi_\ell(r_\ell)$ with Jacobian $A_\ell(x) = \partial\phi_\ell/\partial r_\ell$, let $Q_{\ell,k}(x)$ be an orthonormal basis of the top-k right-singular vectors of $A_\ell(x)$. Two gauge-safe statistics are recorded per stage: the dominant-subspace alignment $a_{\ell,k}(x, x') = k^{-1}\|Q_{\ell,k}(x)^\top Q_{\ell,k}(x')\|_F^2$, equal to one for identical subspaces and decreasing as they rotate, reported within and across event strata; and, for each declared factor h, a contrastive direction $v_{\ell,h} \propto \mathbb{E}[r_\ell \mid h{+}\Delta h] - \mathbb{E}[r_\ell \mid h]$ built from matched interventions, whose alignment with the clean-stratum dominant subspace is the attribution feature. Both depend on subspaces and input–output sensitivities, never on raw coordinates, so they are invariant under the reparameterizations a trained network is free to choose. Together they are the layerwise sensitivity profile of C2. The prediction frozen on Tier 1 before any Tier-2 run is that N-family cells rotate $Q_{\ell,k}$ at the stages nearest the input and place $v_{\ell,\mathrm{N}}$ inside the clean dominant subspace, whereas S-family cells leave early-stage alignment near one and place $v_{\ell,\mathrm{S}}$ outside it. Activation interventions $r_\ell \mapsto r_\ell + \alpha v_{\ell,h}$ then test whether a candidate direction causally moves the intended physics coordinate, which is the patching check that precedes any repair experiment.

4 Claims

C1—Detection. At a 1% false-alert rate on nominal, non-overlapping windows, the best monitoring signature detects at least 80% of held-out abrupt and gradual N/S shifts within five windows, and is non-inferior to the strongest eligible

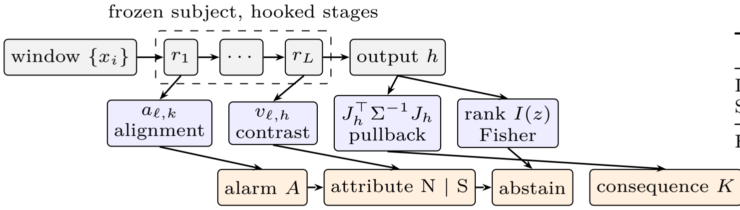


Figure 1: Monitor stack on a frozen subject. Hooked stages feed two gauge-safe layerwise statistics (alignment $a_{\ell,k}$, contrastive direction $v_{\ell,h}$); the output map feeds the pullback metric and the Fisher rank. Alarm, N/S attribution, abstention, and the consequence measure K are separate endpoints with separate false-alert budgets, and every arrow is a pre-registered prediction on Tier 1 before Tier 2 runs.

Table 1: C4 alarm–consequence matrix and endpoints.

	$K < \kappa_m$	$K \geq \kappa_m$
Low alarm	quiet/benign	missed consequence
Strong alarm	harmless alarm	detected consequence
Endpoints	Primary: among strong-alarm cells, AUROC/AUPRC for $K \geq \kappa_m$ and conditional high-consequence risk; also the low-alarm/high-consequence rate. Secondary: within-family $\rho(A, K)$ and the high-alarm/low-consequence rate.	

baseline by a five-point power margin. Delay is primary in *windows*.

C2—Attribution with abstention. Two endpoint families are co-primary. The operational family is macro-F1 over {N,S} on held-out cases, with N+S mixtures scored by multi-label F1. The mechanism family is standardized displacement, neighbour retention, subspace principal angle, and the layerwise sensitivity profile, each normalized by clean within-stratum dispersion.

A split-conformal nonconformity score [4] is used to *abstain* on cases that do not resemble the declared training families. Unseen-family detection is evaluated by AUROC, while attribution quality under abstention is reported through selective risk and macro-F1 as functions of retained coverage [5]. The selective endpoint is risk–coverage AUC, with N/S macro-F1 at 80% retained coverage reported alongside it. Detection of the unknown and attribution among the known are thus not scored as if they were the same quantity.

The paired difference ΔF1 between the full layerwise and all-generic arms is reported with a 95% confidence interval, and read three ways. A practically meaningful benefit requires the interval to exclude zero *and* the point estimate to reach 0.10 macro-F1; equivalence is supported if the interval lies within a predeclared ± 0.05 margin; otherwise the result is inconclusive. This lets a strong generic baseline produce a meaningful negative result — internal hooks may not justify their cost. A mechanism metric is promoted from development to the frozen confirmatory set only if it responds monotonically to the positive-control severity and adds separation beyond the generic arm; otherwise it remains descriptive.

C3—Cost. Ten-percent event sampling plus fixed-size sketches yields p95 replay-path latency overhead $< 10\%$, $\geq 10\times$ less representation storage, and ≤ 5 points of power or F1 loss relative to full capture.

C4—Is an alarm informative about scientific consequence? Let A be the monitor's alarm magnitude and K the downstream scientific loss for the same intervention cell. High consequence is $K \geq \kappa_m$, where κ_m is a testbed-specific scientific tolerance frozen before evaluation — provisionally a 10% degradation for the waveform amplitude task, and for angular or clustering metrics a physically meaningful change in resolution, efficiency, purity, or ARI rather than a generic percentage.

The *controlled simulator* decides this, because the generator supplies both the assumed $\hat{\Sigma}$ used by the encoder and the realized Σ, permitting a controlled test of whether alarms correspond to degradation in the correct likelihood-weighted residual. The public TIDMAD score is a development consequence only: its authors state it lacks direct relevance to fundamental science, so the TIDMAD arm's confirmatory consequence is a relative exclusion-limit comparison on a frozen science-file subset. C4 is primary on the waveform track and secondary on the point-cloud testbed, where consequence is downstream reconstruction error and no exact covariance exists. We state that asymmetry rather than implying symmetric coverage.

Dissociation design. Because alarm magnitude and consequence are both monotone in intervention severity, a pooled alarm–consequence correlation is mechanically positive and evidentially empty. C4 is therefore evaluated within severity strata, and the design includes two perturbation families that decouple the quantities by construction: an *output-null* arm (latent displacement along the local output Jacobian's null space; large alarm, $\approx$ zero consequence) and a norm-matched *output-aligned* arm (same displacement magnitude along the leading right-singular directions; large consequence). A monitor that ranks by unweighted Euclidean displacement fails this pair by construction; the pullback-whitened alarm of Section 3 passes, because the two arms differ precisely in their length under $J_h^\top \Sigma^{-1} J_h$. This converts C4 from a correlation into a designed test with a predicted sign and gives every magnitude-only baseline a failure mode it cannot escape.

5 Testbeds and monitoring design

Tier 2 — neutrino-telescope point clouds (realism: ORACLE-Paired). One primary public configuration with a released checkpoint. The layerwise arm hooks all four DynEdge convolution blocks plus the post-processing and readout stages, not only the 128-dimensional pre-head event representation. The data are our own production: the released NuBench datasets are independent per-geometry productions and share no latent events, so the paired detector-geometry contrast is constructed by re-simulation — Prometheus [16] re-uses one seeded injection across detector geometries (here ORCA and ARCA, both water, so layout is not confounded with medium and the photon stage is exactly reproducible), with the NuBench detector-response emulation reimplemented in-project, which is precisely where the acquisition knobs live. Per geometry, from one injection: module dropout, hit thinning, time jitter, gain drift and noise-rate cells instantiate N; S windows use clean low-density strata in energy, topology, vertex/containment, and multiplicity, estimated from training data only — with the caveat that module loss *lowers* observed multiplicity, so every N cell is matched by clean events at the same observed multiplicity, charge, and time spread; otherwise attribution reduces to recognizing the corruption operator's fingerprint. Undeclared families (double-bang ν_τ, through-going muons, pile-up, correlated module noise) are held entirely outside the declared contract for the abstention endpoint, and the injection identifier is the pairing key across geometries. One secondary architecture serves as a consistency check. Until the paired set exists, the released Hexagon water/ice pair supports only a distribution-level medium comparison, and is labelled as such. No assumed covariance exists in this tier — deliberately: it tests, out of sample, the monitor-failure predictions pre-registered on Tier 1, and it is where the one alarm computable *without* knowing Σ (the Jacobian-projected displacement) earns its deployability claim.

Tier 1 — controlled waveforms (mechanism: ORACLE-Cov). The synthetic arm uses a noise-aware linear control and a generator that constructs PSD/cross-spectral density models, samples seeded frequency-domain coefficients with real-signal symmetry, and adds declared artifacts; the generator reports both the implied and the realized covariance of every ensemble. The inverse-PSD-weighted training objective makes the encoder's assumed covariance $\hat{\Sigma}$ a literal property of the loss, so $\hat{\Sigma} \neq \Sigma$ is a controlled experimental lever, swept through the condition number $\kappa(\hat{\Sigma}^{-1}\Sigma)$. Every theoretical claim is decided here, and the power analysis that sizes the other tiers runs here, because only here is the null exactly simulable.

Tier 3 — TIDMAD (real data). The public TIDMAD [22] arm uses a ~ 0.6M-parameter two-stage domain transformer, comparing otherwise identical MSE and inverse-PSD-weighted objectives — two assumed covariances on identical real electronics noise, the naturally occurring real-data echo of the Tier-1 contrast. The release's denoising score is a development consequence only; the confirmatory consequence is the relative exclusion-limit comparison. This compact transformer is a diagnostic subject, not a foundation model; a larger pretrained model is only a later gated comparison.

Monitoring comparison. Each window records input-quality summaries, named-layer moments or sketches, outputs, uncertainty, delayed task metrics, and calibration. A regularized logistic classifier is fitted with identical splits to five arms: input only; output/uncertainty; final embedding only; all generic-monitor outputs; and the full layerwise signature (the alignment and contrastive-direction statistics of Section 3 at every hooked stage). The fourth is the honest strong baseline, committed to in advance. Eligible baselines — corrected univariate input tests, RBF-MMD [6], a classifier two-sample test [7], embedding mean/covariance distance, output tests, uncertainty/entropy — receive identical windows and false-alert budgets. Clean miscalibration is a model result, not automatically a software bug [8]; latent displacement is normalized by clean within-class dispersion. Held-out severities, seeds, support strata, mixtures, and one unseen family prevent perturbation lookup; windows and thresholds are frozen after development; intervals resample event groups and perturbation seeds separately; null results require a passed positive control, adequate precision, and a predeclared equivalence margin.

Reporting standard. Every learned arm and baseline is reported with its parameter count, training cost, and inference time and memory per event, and arms compared against each other are matched in parameter count, not in tuning effort. Every endpoint is reported inclusively and on the stratum where the methods are predicted to differ (highest severity, lowest SNR, boundary support), under the same metric names on every tier. Every threshold — alarm, attribution, abstention — is reported as a curve at the pre-registered false-alert budget (risk–coverage for abstention), never as a single operating point. Each learned ladder contains an extrapolation arm trained inside and evaluated outside the declared severity and multiplicity range, labelled as such. The hyperparameters of every method appear in one table with the values used.

Table 2: Core experiment matrix. C1–C3 run on both testbeds; C4 is primary on waveforms.

ID	Test	Primary endpoints
E0	Clean replay; deliberate E faults	Reference discrepancy; gate sensitivity/specificity
E1	Held-out abrupt/-gradual N/S	Power at 1% false alert; AUROC/AUPRC; delay in windows
E2	N/S attribution, mixtures, unseen family	Macro/multi-label F1 over {N,S}; paired CI vs generic arm; unknown AUROC
E3	Capture vs sample/sketch	p50/p95 latency; bytes/event; power/F1 loss
E4	Alarm vs consequence (C4)	$\rho(A, K)$; AUROC for $K \geq \kappa_m$; off-diagonal rates
E5	Paired-geometry transfer	Pre-registered Tier-1 failure predictions tested once on ORACLE-Paired under the same metric names (bridge table: within-T_S residual variance under N, along-$T_S^\perp$ displacement under S, abstention AUROC, $\rho(A, K)$); layer-profile stability across geometries

6 Staged joint roadmap

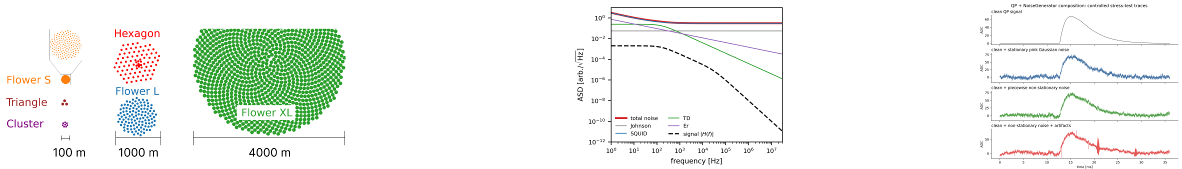
Phase 0 (now): mechanism tier. The output-null/output-aligned dissociation and the $\kappa(\hat{\Sigma}^{-1}\Sigma)$ sweep run on the controlled simulator today — no external dependency, first citable result, and the power analysis that sizes every later phase. Monitor-failure predictions are frozen here before any Tier-2 run.

Phase 1 (B): paired detector geometry. Generate common latent events by Prometheus re-simulation (one seeded injection, two geometries at a common site/depth) under one declared acquisition intervention; the released NuBench sets are not event-paired, so the interim released-data contrast (Hexagon water vs. ice) is reported only as a distribution-level consistency comparison, never a causal layout claim.

Phase 2 (A): Panda scale transfer. On one frozen public Panda checkpoint (pip-installable, auto-downloading data; the cross-modality Panda V2 backbone [18] once its weights are released; SPINE remains the later domain-specific extension), test one physically justified perturbation and one point-representation metric against the released reconstruction endpoints.

Phase 3 (A$\oplus$B): target experiment. If both pilots pass, compare physics-supported geometry/acquisition interventions with matched non-physical representation changes at the same displacement magnitude. This is the unifying attribution test, not an assumption that every learned coordinate is meaningful.

Relation to chained-model uncertainty propaga-



(a) detector geometries (b) detector signal/noise spectrum (c) generated waveform stress tests

Figure 2: Complementary testbeds. (a) Six top-view geometries, reproduced from NuBench Fig. 3 [1]. (b) The analytic Al_2O_3/Al athermal reference budget — component and total noise ASDs with the two-pole signal response — regenerated from the parameterised model in `noise_module.reference_budget`. (c) The generator composes a clean optimal-filter-like pulse with stationary colored noise, piecewise nonstationarity, and explicit artifacts. Panels (b)–(c) are project outputs.

tion. Douglas et al. [3] propagate input uncertainty through a chained neutrino-LAr reconstruction and compare uncertainty-aware and blinded models under synthetic noise. Here the model is frozen: the question is whether its representation attributes a declared failure, abstains outside scope, and links an alarm to consequence. Propagated uncertainty is therefore a candidate covariate, not the endpoint.

Relation to shift monitoring and localization. The components of this study are individually established, and we claim the physics instantiation and the evaluation protocol, not the components. Shift-type identification from a frozen encoder exists in medical imaging [9], building on the acquisition-versus-population taxonomy of [10]; harmful-versus-benign shift detection is a named research line [11, 12], and performance changes have been attributed to specific shift mechanisms [13]. That shift type determines *which layers* to adapt is the headline of surgical fine-tuning [14], and localization-then-edit validation is known to be a weak inference [15], which is why any later repair experiment here is preceded by an activation-patching causal check rather than resting on adaptation gains. What none of these supply, and what this study tests, is (i) a consequence variable that is a physics observable under a known measurement covariance rather than classification accuracy, (ii) the conditional-on-alarm alarm–consequence endpoint of Table 1 with designed alarm/consequence dissociations, and (iii) any monitoring result at all for learned neutrino-telescope reconstruction.

7 Collaboration

The agreed sequence is B→A→A⊕B, while the waveform/TIDMAD arm runs in parallel and supplies the first code-complete contrast. Work is consolidated in the shared ORACLE repository. A proposed fortnightly meeting reviews one frozen protocol or result; initial ownership is waveform infrastructure and generic monitoring on Dowling's side, and geometry/SPINE physics choices and validation on Junjie's side, with thresholds and claim changes approved jointly. The first meeting should freeze one geometry intervention, pairing variables, one representation endpoint, and one downstream physics endpoint.

8 Feasibility gate

Dataset access, checkpoint loading, released-prediction rescoring, intermediate-hook extraction, and a monotone module-dropout positive control have been validated (15.28° → 25.12° median angular error and 10-NN retention 1.000 → 0.393 across 0–50% dropout), with the caveat that the pilot sample was enriched and non-randomly drawn; the audited scripts redraw it uniformly with nested severities and paired intervals. Exact re-inference parity remains unresolved: the current pipeline differs from the released directions by 0.944° median angular error. Two identified confounds are resolved first: the released-metric rescoring had grouped by the track/cascade topology flag while Table 6 categories are interaction classes, and the pilot substitutes five preprocessing methods of the released detector, which is itself a candidate cause and is now disclosed with the outputs. The decisive next test is cheap: identical re-inference on CPU and GPU separates a k-NN backend cause (CPU≠GPU) from preprocessing, alignment, or version skew (CPU=GPU≠release). Exact identity is a software control, separate from the scientific geometry gate. On a fixed development subset we will test whether the residual offset is stable across severity and event strata. Stable offsets permit a predeclared backend tolerance for the paired contrast; severity-dependent drift is a confound and keeps the gate closed.

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